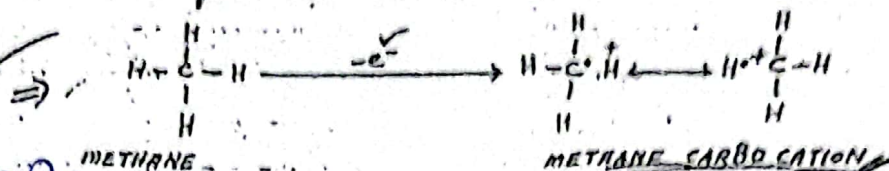


TRIPHENYL AMINE

TRIPHENYL AMINE CATION

In case of methane cation, the reaction is as:



METHANE

METHANE CARBO CATION

DETECTION OF FREE RADICALS

There are different methods available for the detection of free radicals. Some important methods are given in the following for the detection of the stable and unstable free radicals.

(i) MAGNETIC MOMENTS

Associated with the spin of an electron is a magnetic moment, which can be expressed by a quantum number of $+\frac{1}{2}$ or $-\frac{1}{2}$. According to the Pauli Exclusion principle, any two electrons occupying the same orbital must have opposite spin, so the total magnetic moment is zero for any species in which all the electrons are paired. In radical, however, one or more electrons are unpaired, so there is a net magnetic moment and the species are paramagnetic. Radicals can therefore be detected by magnetic measurements, but for this technique, a relatively high concentration of radicals is required. These radicals show paramagnetic behaviour in the applied magnetic field also.

(ii) ELECTRON SPIN RESONANCE [E.S.R.]

A much more important technique is electron spin resonance (E.S.R), also called electron paramagnetic.

resonance (EPR). The principles of ESR is similar to

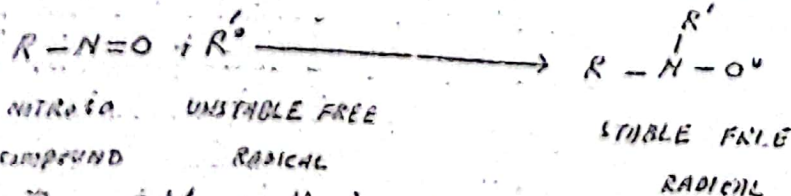
that of NMR, except that electron spin is involved rather than nuclear spin. In this method, a radical is placed in electromagnetic field that provides energy to flip the electrons from lower energy state (orbital) to the higher one, and the radio-frequency signal gives an ESR spectrum. If there are paired electrons, they cancel out the effect of each other due to the opposite spin, so that no peak appears in spectrum. However, in case of unpaired electrons, the spectrum indicates the peak which indicate the presence of free radicals.

Since only free radicals give an ESR spectrum, this method can be used to detect the presence of free radicals and to determine their concentrations. ESR spectra have been observed for radicals with lifetimes considerably less than 1 second.

(iii) SPIN TRAPPING METHOD

Failure to observe an electron spin resonance spectrum does not prove that radicals are not involved, since the concentration may be too low for direct observation. In such cases the spin trapping technique can be used.

In this technique a compound is added that is able to combine with very reactive radical to produce more persistent radicals and the new radicals can be observed by ESR. The most important spin-trapping compounds are nitroso compounds, which react with radicals to give fairly stable nitroxide radicals.

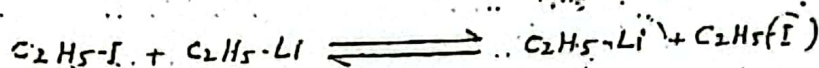


The stable radicals formed are easily detected by electron spin resonance (ESR).

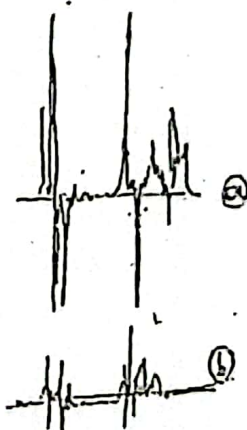
(IV) NUCLEAR MAGNETIC RESONANCE METHOD.

Another magnetic technique for the detection of free radicals is the use of an ordinary nuclear magnetic resonance instrument (nmr). It was discovered that if an nmr spectrum is taken during the course of a reaction, certain signals may be enhanced, either in positive or negative directions, others may be reduced. When this type of behaviour, called "CHEMICALLY INDUCED DYNAMIC NUCLEAR POLARIZATION (CIDNP)" is found in the nmr spectrum of the product of a reaction, it means that at least a portion of that product was formed via the intermediacy of a free radical.

For example, the question was raised whether radicals were intermediates in the exchange reaction between ethyl iodide and ethyl lithium.



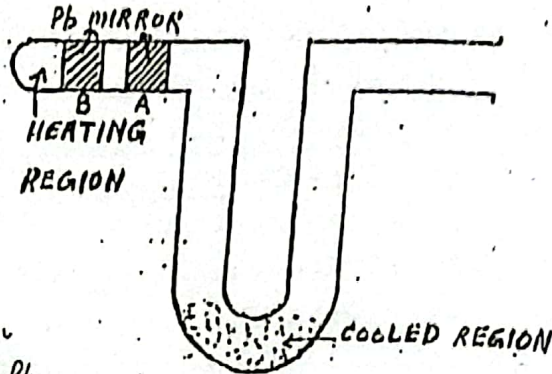
Note that curve "a" shows an nmr spectrum taken during the course of the reaction. Curve "b" is a reference spectrum of ethyl iodide. Note that in curve 'a' some of the ethyl iodide signals are enhanced, others go below the baseline (negative enhancement; also called emission). Thus the ethyl iodide formed in the exchange show CIDNP and hence was formed via free radical intermediate. CIDNP results when protons in a reacting molecule become dynamically coupled to an unpaired electron while traversing the path from reactants to products. Although the presence of CIDNP almost always means that a free radical is involved, but its absence does not prove that a free radical intermediate is necessarily absent.



(V) PERCH. METHOD

This method is also used to detect the stable and unstable free radicals. In this method, the detection of free radicals is done by the formation of mirror.

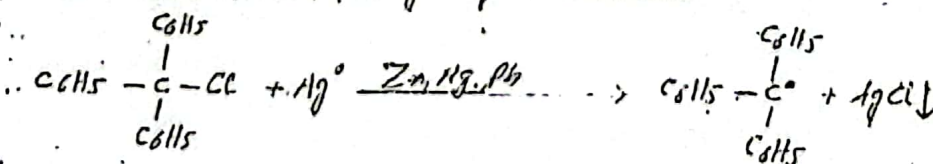
For example, for the detection of methyl radical $\text{CH}_3^\bullet$, the tube shown in the fig. is used. When tetramethyl lead $(\text{CH}_3)_4\text{Pb}$ is placed in this tube and heated at the point B, the compound breaks to give $\text{CH}_3^\bullet$ radical and free Pb, and this lead is deposited in the form of mirror inside the tube. Now if heating is stopped at point A, $\text{CH}_3^\bullet$ radical and Pb recombine to form $(\text{CH}_3)_4\text{Pb}$ once again.



Now, when the compound is heated again at point B, the methyl radicals are formed and the mirror reappears at point B. But when the heating is stopped, again Pb and $\text{CH}_3^\bullet$ radical combine to form $(\text{CH}_3)_4\text{Pb}$ in the cooling zone of the tube.

(VI) GOMBERG METHOD

This method was developed by Gomberg in 1900. In this method stable radicals can be detected by the production of coloured solution. Gomberg used silver and lead and got free radicals.



TRIPHENYL METHYL CHLORIDE

TRIPHENYL METHYL RADICAL

The free radicals formed were detected by the decolouration of iodine solution.

